

Applied Machine Learning

Evaluation of Classifiers and Regressors



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CHALMERS

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evaluating classifiers and regressors: typical setup

- ▶ **intrinsic** evaluation: we want to see how good the outputs are
- ▶ we do this by **comparing** to a gold standard

evaluating classification systems

main idea when evaluating classifiers

- is the output correct or not?

<i>gold standard</i>	<i>predicted</i>
cat	dog
dog	dog
dog	dog
dog	dog
dog	dog
dog	dog
dog	dog
dog	cat
dog	dog
dog	dog

accuracy and error rate

$$\text{accuracy} = \frac{\text{nbr. correct}}{N}$$

$$\text{error rate} = \frac{\text{nbr. incorrect}}{N}$$

confusion matrix

		Actual class		
		Cat	Dog	Rabbit
Predicted class	Cat	5	2	0
	Dog	3	3	2
	Rabbit	0	1	11

[[source](#)]

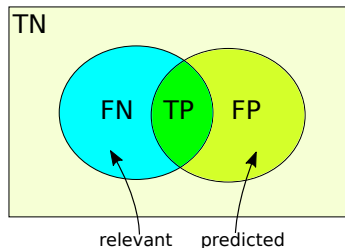
considering one class

- ▶ let's say we're interested in one class in particular
 - ▶ “spotting problems” or “finding the needles in the haystack”

		Actual class	
		Cat	Non-cat
Predicted class	Cat	5 True Positives	2 False Positives
	Non-cat	3 False Negatives	17 True Negatives

evaluation metrics derived from a confusion matrix

- ▶ several evaluation metrics are defined from the cells in a confusion matrix
 - ▶ depending on the scientific field
- ▶ these metrics often come in pairs
 - ▶ overgeneration vs undergeneration



see https://en.wikipedia.org/wiki/Confusion_matrix

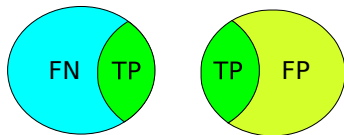
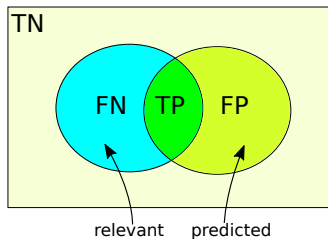
precision and recall

- ▶ the **precision** and **recall** are commonly used for evaluating classifiers

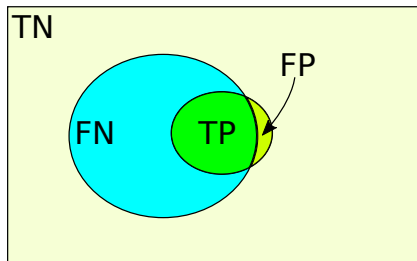
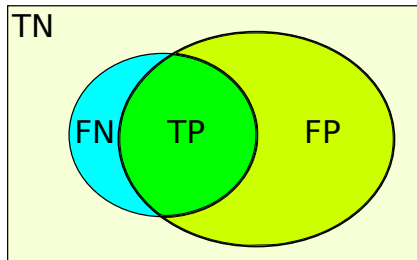
$$P = \frac{TP}{TP + FP} \quad R = \frac{TP}{TP + FN}$$

- ▶ in scikit-learn:

- ▶ `sklearn.metrics.precision_score`
- ▶ `sklearn.metrics.recall_score`
- ▶ `sklearn.metrics.classification_report`



precision / recall tradeoff



F-score

- ▶ the **F-score** is the harmonic mean of the precision and recall

$$F = \frac{2 \cdot P \cdot R}{P + R}$$

- ▶ in scikit-learn: `sklearn.metrics.f1_score`

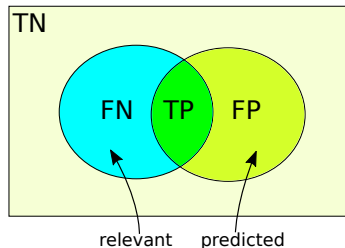
other similar pairs of metrics

- **sensitivity/specificity** (in medicine)

$$\text{Sen} = \frac{TP}{TP + FN} \quad \text{Spe} = \frac{TN}{TN + FP}$$

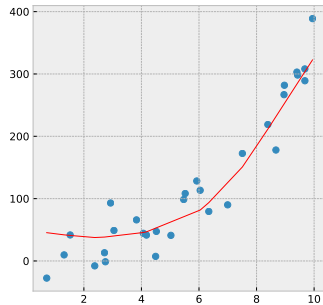
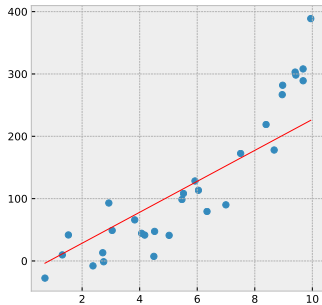
- **true positive rate/false positive rate**

$$\text{TPR} = \frac{TP}{TP + FN} \quad \text{FPR} = \frac{FP}{FP + TN}$$

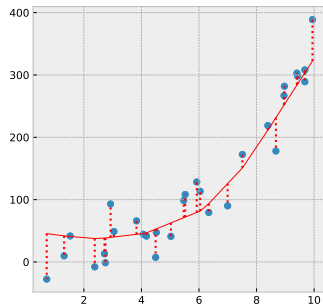
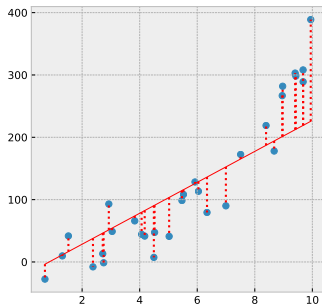


evaluating regression systems

example



errors



mean squared error and mean absolute error

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

(N = number of instances, $\hat{y}_i$ = prediction for instance i)

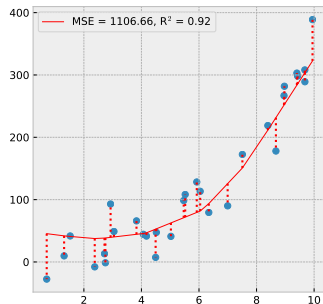
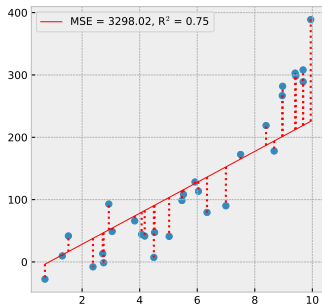
R^2 : the coefficient of determination

- ▶ the MSE and MAE scores depend on the scale
- ▶ they don't directly tell whether the regressor behaves “well”
- ▶ the **coefficient of determination** (R^2) is an evaluation score that does not depend on the scale

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

- ▶ in case of a perfect regression, this score is 1
- ▶ if completely messed up, 0 or negative

MSE and R^2 for the example



what to do about ordinal scales



Grekisk pizzeria
Plaka
Utlandagatan 14

Det är inte mycket som lockar till en pizzeria i dag eftersom det är oftast är bukfylla utan vidare ambitioner, men ibland finns en parallell taffel som kan glimra i hemlighet. Vi gick till Plaka som ligger i ett annars restaurangsnålt område.

Vi anade snart att det var servitrisen som var kvällens blyga behållning – snabb, vänlig, lyhörd. Plaka har en pizzameny med hela 49 varianter mellan 85 och 95 kronor som slutar med en Gyros Pizza. Det föreföll lika äckligt som Kebab Pizza. En maträtt från helvetet som kanske borde returneras dit.

Därför sneglade vi mot den grekiska menyn med den famlande tanken om att den grekiska krisen kanske ändå inte nått ända dit. Vi beställde två standardförrätter. Grillad halloumi och friterad bläckfisk. I halvfabrikatens värld fordras inget geni direkt

- ▶ see e.g. Baccianella et al. (2009) *Evaluation Measures for Ordinal Regression*

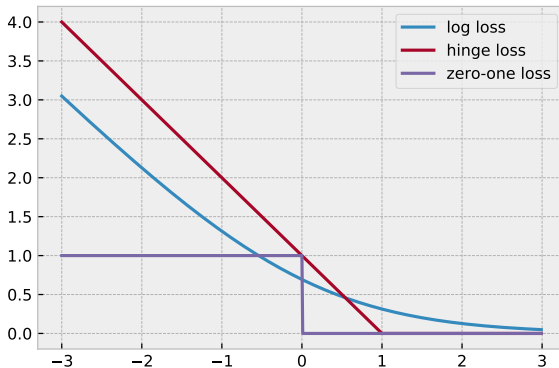
detour: training losses and evaluation metrics

regression and classification evaluation

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad \text{error} = \frac{1}{N} \sum_{i=1}^N 1[y_i \neq \hat{y}_i]$$

- ▶ for **regression**, MSE is often used
 - ▶ as the **quality metric** when we **evaluate**
 - ▶ as the **loss function** when we **train**
- ▶ but typically, for **classification** what we compute during training and testing is different
 - ▶ such as **hinge loss or log loss during training**

loss functions



summary

- ▶ most metrics are based on comparing to a **gold standard**
- ▶ for **classification**, various metrics based on whether the output was correct or not
 - ▶ straightforwardly: **accuracy** and error rate
 - ▶ positives and negatives: **precision** and **recall**
- ▶ for **regression**, we measure the differences between predicted and desired values
 - ▶ **MSE, MAE, R^2**
- ▶ next: more about evaluating classifiers!